

● UIPATH HACKATHON 2026

VendorTrust AI

End-to-End Third-Party Risk Assessment Pipeline

AI-driven vendor onboarding — risk scoring, fraud detection, parallel compliance checks, and multi-tier approvals — all orchestrated with UiPath Maestro BPMN.

⚙️ Maestro BPMN

🤖 AI Agents

📄 Document Understanding

✅ Action Center

🔗 ERP Integration

⚠ Vendor Onboarding is Broken

Manual processes are introducing severe risk and holding back business growth.



🕒 2–4 Weeks Per Vendor

Slow processes relying on manual review, email chains, and isolated siloed approvals.



⚠ 60% of Compliance Breaches

The majority of serious security compliance incidents can be traced to inadequate third-party vetting.



💰 \$4.4M Average Breach Cost

Third-party breaches require complex regulatory containment, escalating associated costs.



Frustrated executive handling multiple paper files

Uncoordinated parallel checks (AML, sanctions, credit, legal) performed in isolation result in **73% of enterprises** being impacted by vendor risks.

VendorTrust AI — Risk Intelligence

10× Faster Onboarding

Automated 6-phase pipeline to quickly assess, approve, or flag any third-party vendor in mere hours rather than weeks.

AI Risk Intelligence

Multi-model AI framework scoring real-time international sanctions, credit data, forgery, and overall financial health.

End-to-End Orchestration

Maestro BPMN orchestrates every stage, seamlessly driving flow from initial web intake to direct ERP system creation.

Smart Human-in-Loop

Action Center engages specialized compliance experts only when confidence is low, optimizing labor costs.

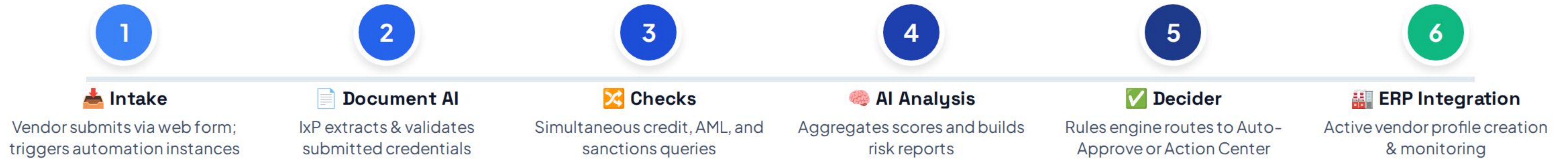
Document AI Extraction

Highly accurate extraction and verification of vendor compliance documentation at 95%+ precision rate.

ERP & Continuous Monitor

Approved vendors are automatically synchronized with central ERP records alongside continuous hazard alerts.

6-Phase Orchestration Pipeline



 UiPath Maestro BPMN engine coordinates and drives the complete end-to-end flow.

 Real-time parallel splits

Powered by Multi-Model AI

Document AI Layer

- ✓ Automatic document classification (Tax, Registry, Certification)
- ✓ Field extraction (Tax ID, registration date, legal entities)
- ✓ Underlying document completeness & authenticity verification

Risk Scoring Engine

- ✓ Real-time database match against Sanctions & PEP registries
- ✓ Instant credit API score pulling (0 - 100 risk score matrix)
- ✓ Flexible and adjustable threshold parameters

Fraud Detection

- ✓ Digital document forgery pattern monitoring
- ✓ Detecting hidden shell company and holding hierarchies
- ✓ Dynamic cross-comparison of regional address inconsistencies

AI Risk Summary Generation

- ✓ Automated natural language summaries generated via LLM Gateway
- ✓ Actionable decision rationale with verifiable evidence
- ✓ Audit-ready documentation stored automatically in compliance database

Built on the UiPath Agentic Platform

Layer	Components & Applied Technologies
 Orchestration	UiPath Maestro BPMN (ProcessOrchestration) · Over 70 modeling elements · 6 core logical phases · Dual parallel gateways with conditional paths
 AI & Automation	UiPath Document Understanding (IxP engine) · Global LLM Gateway integration · AI Center secure custom risk ranking agents
 Human-in-the-Loop	UiPath Action Center portal · Custom 3-tier escalations (Officer -> Manager -> Legal) mapped dynamically against aggregate scores
 Integrations & APIs	Sanctions API verification · Credit Bureau data pulling · AML/KYC background tools · Direct SAP/ERP Master Data API · Enterprise notification engines

BPMN Deep Dive — Key Patterns

Parallel Risk Gateway

Splits into four parallel microservice queries simultaneously, recombining via a strict "And Join" to wait for all risk factor feedback before finalizing overall calculations.

Parallel Split

Parallel Join

Risk-Based Routing

Dynamically routes processes to specific Action Center task cues: scores <30 auto-approve, scores 30–70 escalate to manual triage review, scores >70 trigger Legal.

Exclusive XOR

Score Variables

SLA-Aware Timers

Boundary task timers monitor Action Center processing. Triggers system-wide warning signals or auto-escalates vendor review assignments upon exceeding default thresholds.

Timer Boundary

Error Events

Resubmission Loops

If document validation reports gaps, a targeted feedback branch alerts vendor point of contact to upload correcting documents (restricted to a maximum of 3 loops).

Loop Gateway

Message Events

Demo Run — TechSupply Solutions Ltd

INPUT PAYLOAD SCHEMA

```
{
  vendorName: "TechSupply Solutions",
  vendorId: "VA-2026-001234",
  contact: "Sarah Mitchell",
  category: "IT Services & Cloud",
  country: "United States",
  revenue: 12500000,
  yearsInBusiness: 8
}
```

 Low risk classification automates final system approvals.

LIVE EXECUTION WALKTHROUGH

STEP 1 Registration Submission: Forms intake instantly fires the Maestro BPMN engine instance.

STEP 2 Document AI Verification: Auto-extracts registration data, verifying certificate validity.

STEP 3 Parallel Sanctions Check: 4 simultaneous external queries pass with no warnings flagged.

SCORE 24 / 100 Auto-Approval: Confirms low risk profile; writes immediately to active SAP master file.

Measurable Business Impact

10x

FASTER ONBOARDING

\$2.1 Million

Estimated savings per 500 vendors



Exceptional Operational Velocity

Shrinks traditional process time from weeks down to hours. Automated parallel paths remove scheduling queue bottlenecks.



Proactive Compliance Defense

Ensures full background check completion on 100% of registrations. Automatically logs immutable audit logs for regulators.



Limitless Process Scaling

Enables easy scaling to accommodate hundreds of onboarding submissions monthly at zero incremental licensing or resource expense.

> Built in 48 Hours

Deliverables Built

- ✓ Complex 70+ element BPMN model
- ✓ Full 6-phase validation pipeline
- ✓ Document AI OCR parsing layer

Hurdles Overcome

Mastered Maestro's parallel join synchronization constraints and structured responsive, multi-tier Action Center routes dynamically.

DEVELOPMENT MILESTONES

0h Ideation & System Archetype Design

Mapped risk scoring algorithms and structured overall orchestration logic steps.

6h Maestro Process Architecture modeling

Established baseline logical pathways, parallel flows, and routing variables.

18h AI Engine Integrations

Constructed custom Document AI extraction rules and LLM validation calls.

48h Deployment & Live Submission Ready ✓

Conducted end-to-end trace debugging runs on physical server nodes.

Roadmap to Production

PHASE1 — CURRENT

6-Phase Pipeline

- ✓ Onboarding flow baseline
- ✓ AI risk analysis & scoring
- ✓ Action Center escalation routing
- ✓ Automatic ERP profile creation

PHASE2 — 3 MONTHS

Continuous Monitoring

- Real-time global sanctions streaming
- Environmental & ESG tracking
- Interactive partner dashboards
- Unified vendor portal front-end

PHASE3 — 12 MONTHS

Predictive Modeling

- Predictive vendor bankruptcy warning
- AI contract risk evaluation
- Tailored industry metric sets
- Marketplace deployment options

Thank You

Making third-party risk management intelligent, automated, and auditable.

Submitted by **Prajyot Birajdar** — UiPath Hackathon 2026. Built with Maestro
ProcessOrchestration BPMN utilizing 70+ logical steps and robust parallel assessment.

SOLUTION ID
3f635e25-6924...

CONTACT INFO
prajyotbirajadar1998
@gmail.com

CATEGORY
Process Orchestration

Image Sources



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